

8th International Flame Chemistry Workshop

25th-26th July 2026

Agenda

<i>Day 1 - Saturday 25th July – Room 205 (second floor)</i>			
	<i>Moderator</i>	8.00-8.30	Registration
Joint Session RCM and Flame Chemistry workshops	<i>Matteo Pelucchi Scott Goldsborough</i>		
		8.30-8.40	Introduction
		8.40-9.05	“Community-driven Collaboration: Scientific Advances via the RCM Workshop Initiatives” Scott Goldsborough, Argonne National Laboratory, USA
		9.05-9.20	“Highlights from the 7 th Flame Chemistry Workshop: challenges and objectives in chemical kinetics modeling of combustion and plasma combustion” Matteo Pelucchi, Politecnico di Milano, Italy
		9.20-9.35	“Combining Theory, Modeling, and Experiment: An Example from DME Research” Markus Kohler – German Aerospace Center, Stuttgart, Germany
		9.35-10.00	Open Discussion
		10.00-10.30	Coffee Break and group picture
Opening FCWS	<i>Matteo Pelucchi Bin Yang</i>	10.30-10.45	Welcome, summary and agenda
Topic 1: Sustainable fuels combustion (LFS, IDT, pollutants): H ₂ , NH ₃ , SAFs	<i>Matteo Pelucchi Liming Cai</i>		
		10.45-11.30	“Theoretical kinetics of reactions between molecular oxygen and nitrogen centered radicals formed during alkylamine oxidation” Yusuyuki Sakai, Ibaraki University, Japan
			“Multiscale physics-based, data-driven studies of combustion and emissions of hydrogen and Ammonia” Mike Burke, Columbia University, USA
			“Combustion chemistry of SAF and supercritical autoignition of H ₂ ” Song Cheng, Hong Kong Polytechnic University, Hong Kong
		11.30-12.15	Open Discussion
		12.15-13.30	Lunch
Topic 2: Advanced diagnostics for	<i>Brandon Rotavera</i>		
		13.30-14.15	“Simultaneous multi-species sensing in shock tubes through variable-pathlength laser

combustion measurements	<i>Andrea Comandini</i>		<i>absorption</i> ” Ron Hanson, Stanford University, USA
			“Synchrotron-based diagnostics for chemical kinetic experiments” Tina Kasper, Paderborn University, Germany
			“Structured combustion data and databases for kinetic mechanism development and validation” Tibor Nagy, Research Centre for Natural Sciences, Hungary
		14.15-15.00	Open Discussion
		15.00-15.30	Coffee Break
Topic 3: Plasma combustion: experiments and modelling	<i>Bin Yang</i>	15.30-16.15	“Computation of the low-energy electron scattering cross section for NH ₃ plasma” Xianwu Jiang, Wuhan University of Technology, China
			“Laser diagnostics of hydrogen: from quadrupole absorption to Raman scattering” Wei Ren, The Chinese University of Hong Kong, Hong Kong
			“Nanosecond-Duration Pulsed Discharges (ns-PD) to Enhance Ignition for Next-Generation Combustion Architectures” Nicholas Tsolas, Auburn University, USA
		16.15-17.00	Open Discussion
Wrap-up	<i>Organizers</i>	17.00-17.30	Summary from Day 1
Poster reception		17.30-19.30	Poster Session (Room 314 & 315), reception

<i>Day 2 - Sunday 26th July - Room 205 (second floor)</i>			
	<i>Moderator</i>	8.00-8.50	Registration
	<i>Matteo Pelucchi</i>	8.50-9.00	Welcome, summary and agenda
Topic 4: AI-TST-ME: automation, benchmarking and knowledge transfer	<i>Feng Zhang</i>	9.00-9.45	“Edge migration of aromatic rings by radical reactions—kinetics and directionality: can the outcome of an aromatic edge migration be predicted on the fly?” Alexander Mebel, Florida International University, USA
			“Transforming combustion chemistry workflows to handle the big data deluge” William H. Green, Massachusetts Institute of Technology, USA
			“Automated kinetic data generation for combustion models using the tabulated model of transition states (TMTS) method” Baptiste Sirjean, Centre National de la Recherche Scientifique, France
		9.45-10.30	Open Discussion
		10.30-11.00	Coffee Break

Topic 5: Chemical Kinetics Mechanisms: Progress in accuracy and details.	<i>Brandon Rotavera</i>	11.00-11.45	<i>“Details in elementary kinetics: setting the foundations for combustion kinetics modeling”</i> Raghu Sivaramakrishnan, Argonne National Laboratory, USA
			<i>“Speciation experiments in kinetics: what do we learn from modeling?”</i> Guillaume Dayma, Centre National de la Recherche Scientifique, France
			<i>“Developing detailed chemical kinetic mechanisms for fuel combustion”</i> Henry Curran, University of Galway, Ireland
		11.45-12.30	Open Discussion
		12.30-13.45	Lunch
Discussion and actions	<i>Matteo Pelucchi Bin Yang</i>	13.45-15.30	- Open discussion on challenges - Feasible joint actions and initiatives for Topic 1-5 - Feedback and suggestions from participants - ...